

Scenario 3: Dynamic Residual Risk with Learned Common Gamma

Detailed MSSTNB simulation validation report

2026-05-27

Table of contents

1	Overview	1
2	Scenario design	2
3	Simulation and fitting settings	3
4	Relationship to Scenario 2	4
5	Covariate and latent effect separation	6
6	Regression parameter recovery	7
7	Learning the common gamma parameter	8
8	Dynamic residual-risk path recovery	11
9	Spatial and dispersion recovery	12
10	MCMC diagnostics	13
11	Summary and interpretation	15

1 Overview

Scenario 3 evaluates the full dynamic version of the MSSTNB simulation study in which the temporal discount parameter is learned rather than fixed. It is designed as the direct

counterpart to Scenario 2.

Scenario 2 used the same dynamic residual-risk data-generating mechanism but fixed the temporal discount factor at the true value, $\gamma = 0.80$. Scenario 3 keeps the same data-generating mechanism and the same dynamic latent path structure, but it treats γ as unknown and estimates it within the MCMC algorithm.

The central question is therefore:

If the dynamic residual-risk process is present and the common temporal discount parameter is unknown, can the model still recover the regression effects, spatial effects, dispersion parameters, and latent dynamic risk paths?

This question is different from asking whether γ can be estimated perfectly. The discount factor is a temporal smoothing parameter, and previous diagnostics indicated that it is more weakly identified than the regression parameters. The practical success criterion for Scenario 3 is therefore whether learning γ materially harms the main inferential targets.

The main conclusion is positive. Across 20 formal replicates, Scenario 3 produces regression recovery that is nearly identical to Scenario 2. The RMSE biases for $(\beta_0, \beta_1, \beta_2)$ are 0.0084, 0.0121, 0.0084, and the corresponding coverage rates are 1.00, 0.95, 1.00. The learned common γ has average posterior mean 0.813 when the truth is 0.80. The average gamma bias is 0.013, and the average gamma acceptance rate is 0.345.

The dynamic residual-risk recovery is also stable. The average log-lambda RMSE is 0.0477, the average 95% log-lambda coverage is 0.930, and the average correlation between posterior mean and true log λ is 0.786.

2 Scenario design

For region $j = 1, \dots, n_1$ and time $t = 1, \dots, T$, the coarsest-level observation model is

$$Y_{tj} \sim \text{NB}(\mu_{tj}, r_j),$$

with mean

$$\mu_{tj} = e_{tj} \lambda_{tj} \exp\{\beta_0 + \beta_1 x_{1,tj} + \beta_2 x_{2,tj} + \phi_j\}.$$

Here e_{tj} is the offset, $\beta_0, \beta_1, \beta_2$ are regression parameters, ϕ_j is the spatial random effect, r_j is the negative-binomial dispersion parameter, and λ_{tj} is the dynamic residual-risk path.

The difference between Scenario 2 and Scenario 3 is the treatment of γ :

Scenario 2: γ fixed at $\gamma_{\text{truth}} = 0.80$,

Scenario 3: γ learned inside the MCMC.

Scenario 3 uses a common temporal discount parameter,

$$\gamma_1 = \gamma_2 = \dots = \gamma_{n_1} = \gamma.$$

This is intentional. The data-generating mechanism uses a common γ , so the fitted Scenario 3 model also uses a common γ . This makes the comparison with Scenario 2 clean. It also allows all regions to contribute information to one temporal persistence parameter, instead of trying to estimate separate weakly identified γ_j values for each region.

The MCMC updates β , ϕ , τ_ϕ , r , κ , γ , and λ . The δ and ω updates are disabled in this validation scenario. The purpose is to isolate the effect of learning the common temporal discount parameter while keeping the rest of the dynamic model matched to the data-generating mechanism.

3 Simulation and fitting settings

Table 1: Scenario 3 simulation and fitting settings.

Setting	Value
Scenario ID	S3_DYNAMIC_LEARNED_GAMMA_T100
Shared data source	DGP_DYNAMIC_T100
Number of replicates	20
Number of time points	100
Number of regions	9
Gamma truth	0.80
Gamma in fitting model	Learned common gamma
x2 mode	continuous_time
MCMC iterations	40000
Burn in	20000
Thinning	5

Scenario 3 uses the shared source data folder DGP_DYNAMIC_T100. This is important for interpretation. Scenario 2 and Scenario 3 are not two independent simulated datasets. They are

two fitting strategies applied to the same dynamic data-generating mechanism. The purpose is to isolate the effect of learning γ .

The five key design points are:

1. Scenario 2 and Scenario 3 use the same dynamic T100 data source.
2. The main model difference is fixed γ versus learned γ .
3. Scenario 3 uses a learned common γ because the true data-generating mechanism uses a common γ .
4. The main success criterion is whether learning γ preserves recovery of β , ϕ , r , and λ .
5. Gamma recovery is interpreted as a weak-identification diagnostic, not as the sole measure of success.

The use of `continuous_time` for x_2 is also important. A region-level or time-constant x_2 can be accidentally aligned with spatial effects in a small lattice. The formal setting uses a time-varying continuous x_2 to reduce artificial covariate-spatial confounding.

4 Relationship to Scenario 2

The cleanest way to read Scenario 3 is as a direct sensitivity analysis relative to Scenario 2. Scenario 2 asks whether dynamic latent paths can be recovered when temporal smoothing is known. Scenario 3 asks whether similar recovery remains possible when the common temporal smoothing parameter is learned.

Table 2: Conceptual comparison between Scenario 2 and Scenario 3.

Feature	Scenario_2	Scenario_3
Data-generating mechanism	Dynamic residual risk	Dynamic residual risk
Data source	DGP_DYNAMIC_T100	DGP_DYNAMIC_T100
Residual-risk path	Sampled	Sampled
Gamma truth	0.80	0.80
Gamma in fitting	Fixed at truth	Learned by MCMC
Gamma structure	Common gamma	Learned common gamma
Primary validation target	Dynamic path recovery under known smoothing	Robustness of main inference when smoothing is unknown

Table 3: Formal comparison between Scenario 2 and Scenario 3.

metric	S2_fixed_gamma	S3_learned_gamma
beta0_rmse_bias	0.0083	0.0084
beta1_rmse_bias	0.0117	0.0121
beta2_rmse_bias	0.0082	0.0084
beta0_coverage	1.0000	1.0000
beta1_coverage	0.9500	0.9500
beta2_coverage	1.0000	1.0000
phi_rmse	0.0251	0.0251
phi_cor	0.9968	0.9968
r_bias	0.4670	0.4452
log_lambda_rmse	0.0469	0.0477
log_lambda_coverage_95	0.9495	0.9301
cor_log_lambda	0.7865	0.7855
cor_delta_log_lambda	0.1875	0.1847
phi_accept_rate	0.2546	0.2536
r_accept_rate	0.3048	0.3053

The formal comparison shows that learning γ does not materially degrade the primary inference. For example, the β_1 RMSE bias changes from 0.0117 under fixed γ to 0.0121 under learned γ . The corresponding β_1 coverage is 0.95 for Scenario 2 and 0.95 for Scenario 3.

For the latent dynamic path, the log-lambda RMSE is 0.0469 in Scenario 2 and 0.0477 in Scenario 3. The correlation between posterior mean and true $\log \lambda$ is essentially unchanged: 0.786 versus 0.786.

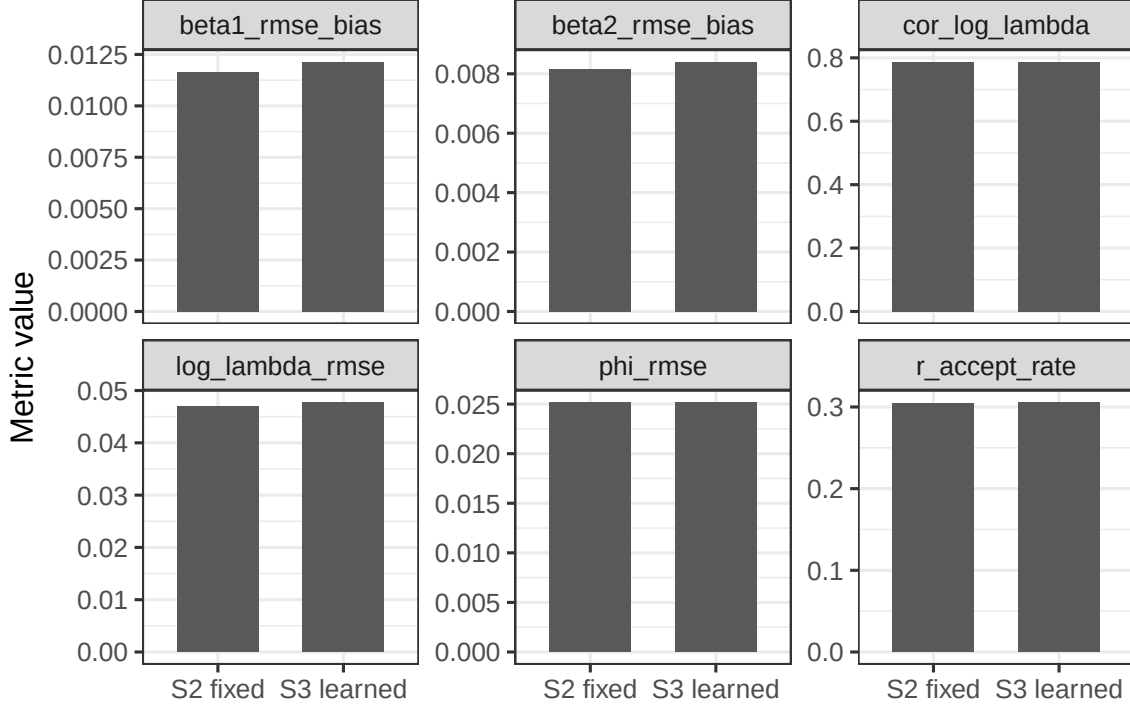


Figure 1: Scenario 2 versus Scenario 3 comparison for selected validation metrics.

This figure is intended to emphasize scale and practical difference, not statistical testing. The bars for Scenario 2 and Scenario 3 are nearly the same for the most important quantities. This supports the interpretation that learned common γ does not meaningfully harm the main posterior recovery targets.

5 Covariate and latent effect separation

Before interpreting regression recovery, it is useful to check whether the covariates are accidentally aligned with the latent components. The mean model contains

$$\beta_2 x_{2,tj} + \phi_j + \log \lambda_{tj},$$

so poor covariate design could create apparent regression bias even if the sampler is correct. This is especially important in a small $n_1 = 9$ spatial lattice.

Table 4: Compact Scenario 3 covariate and latent-effect separation diagnostics. The full replicate-level table is saved in the analysis folder.

diagnostic	mean	sd	min	max
sd_x1	1.0000	0.0000	1.0000	1.0000
sd_x2	1.0000	0.0000	1.0000	1.0000
cor_x1_x2	0.0018	0.0449	-0.0842	0.0680
cor_x2_loglambda	0.0007	0.0541	-0.0881	0.1260
cor_x1_loglambda	0.0211	0.1044	-0.2332	0.2273
cor_x2_phi	0.0021	0.0599	-0.1463	0.1191
cor_x1_phi	-0.0257	0.0790	-0.1888	0.1096
sd_beta2_x2	0.4000	0.0000	0.4000	0.4000

The table is intentionally summarized rather than printed replicate by replicate. The main quantities of interest are the empirical correlations between the covariates and the latent components. Small averages and moderate ranges indicate that regression recovery is not being driven by an artificial alignment between x_2 , ϕ , and $\log \lambda$. The formal Scenario 3 design uses continuous-time x_2 precisely to avoid the region-level confounding pattern that appeared in earlier pilot runs.

6 Regression parameter recovery

Regression recovery is evaluated against the replicate-specific identified truth. The intercept truth varies across replicates because the model uses a recentering convention to separate the regression intercept, spatial effect, and latent residual-risk scale. Therefore, β_0 must be compared to the identified truth stored in the simulation summary.

Table 5: Regression parameter recovery across formal Scenario 3 replicates.

param	truth	post_mean	bias	rmse	cover	post_sd
beta0	-1.5295	-1.5298	-0.0003	0.0084	1.00	0.0093
beta1	0.5000	0.5033	0.0033	0.0121	0.95	0.0102
beta2	-0.4000	-0.4016	-0.0016	0.0084	1.00	0.0097

Regression recovery is strong. The RMSE bias is 0.0121 for β_1 and 0.0084 for β_2 . The coverage rates are 0.95 and 1.00, respectively.

These results are important because they show that learning the temporal discount parameter does not introduce meaningful regression bias. In particular, the learned-gamma model pre-

serves the recovery of both the smooth time-varying covariate effect and the second covariate effect.

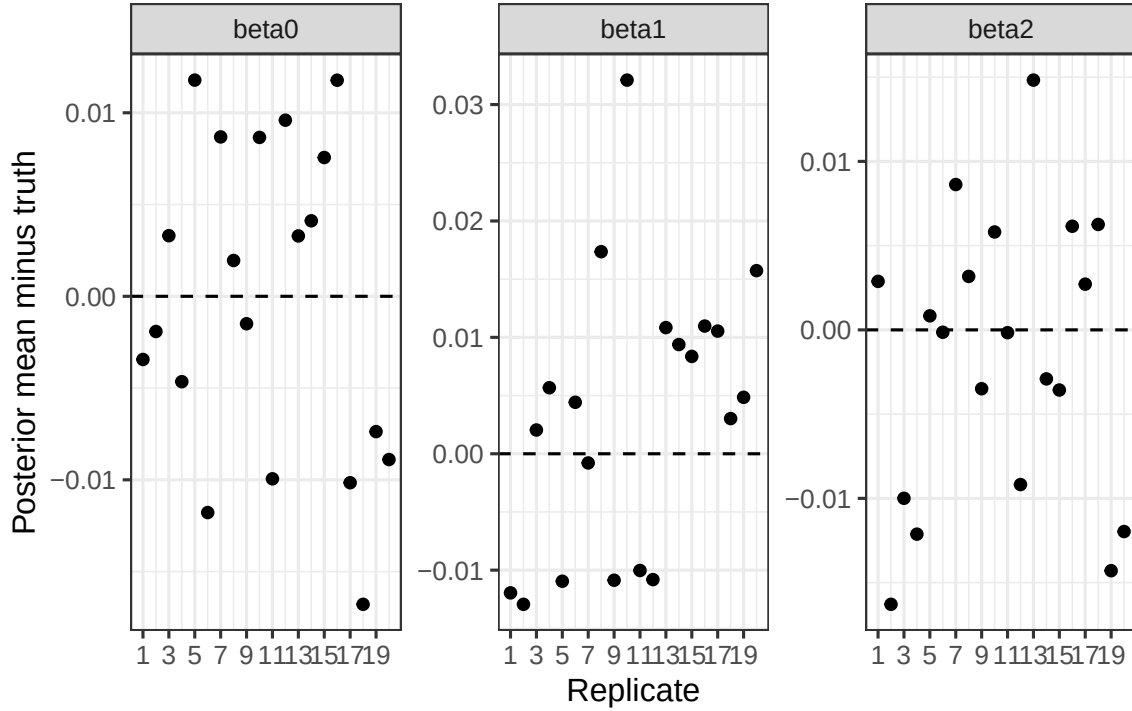


Figure 2: Regression bias by replicate under Scenario 3.

The bias plot should be read as a replicate-level stability diagnostic. Most points are close to zero, and no parameter shows systematic drift in one direction. This is the most important evidence that learning γ does not contaminate the regression block.

7 Learning the common gamma parameter

This is the Scenario 3 specific component. In contrast to Scenario 2, the common temporal discount parameter is updated inside the MCMC. Because the truth is common across regions, the fitted model also uses a common γ .

Table 6: Learned common gamma summary across formal Scenario 3 replicates.

metric	value
Truth gamma	0.8000
Mean posterior gamma	0.8127

metric	value
SD of posterior means across replicates	0.0486
Mean gamma bias	0.0127
RMSE of gamma bias	0.0491
Mean absolute gamma bias	0.0415
Gamma 95% coverage	0.9000
Mean gamma acceptance rate	0.3450

The learned γ is centered close to the truth. The average posterior mean is 0.813, compared with truth 0.80. The average bias is 0.013, and the RMSE of the bias is 0.049. The 95% gamma coverage is 0.90.

This coverage is slightly below the nominal 0.95 target. This slight undercoverage is not interpreted as evidence of model failure. The discount parameter controls temporal smoothing and is less directly identified than the regression coefficients. The more important result is that gamma learning remains stable and does not degrade the main inferential targets. The average gamma acceptance rate is 0.345, which indicates healthy Metropolis-Hastings tuning.

Table 7: Compact replicate-level learned gamma summaries. The posterior median and posterior SD are saved in the CSV output but omitted here to keep the table readable.

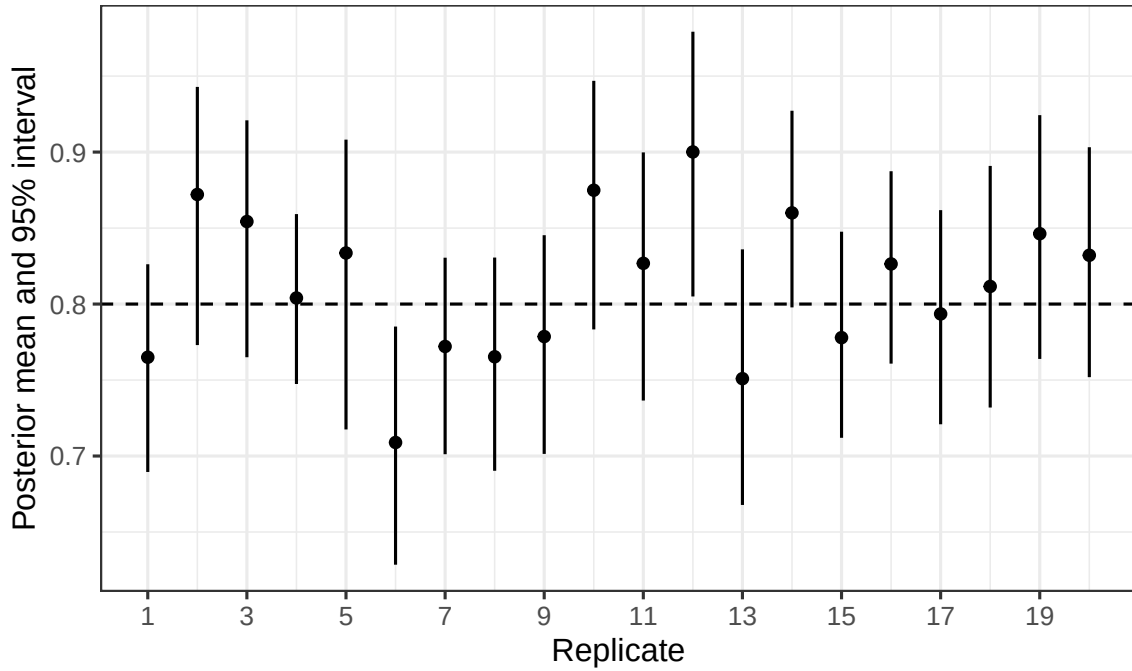
rep	mean	q025	q975	bias	cover	accept
1	0.7650	0.6895	0.8262	-0.0350	1	0.3589
2	0.8721	0.7730	0.9429	0.0721	1	0.3365
3	0.8543	0.7650	0.9209	0.0543	1	0.3320
4	0.8040	0.7474	0.8592	0.0040	1	0.3379
5	0.8336	0.7175	0.9082	0.0336	1	0.3457
6	0.7089	0.6285	0.7852	-0.0911	0	0.3598
7	0.7721	0.7012	0.8305	-0.0279	1	0.3671
8	0.7653	0.6903	0.8306	-0.0347	1	0.3544
9	0.7786	0.7014	0.8453	-0.0214	1	0.3432
10	0.8749	0.7833	0.9469	0.0749	1	0.3204
11	0.8268	0.7365	0.8998	0.0268	1	0.3332
12	0.9001	0.8050	0.9792	0.1001	0	0.3436
13	0.7509	0.6678	0.8360	-0.0491	1	0.3777
14	0.8600	0.7978	0.9272	0.0600	1	0.3595
15	0.7779	0.7120	0.8476	-0.0221	1	0.3396
16	0.8264	0.7608	0.8874	0.0264	1	0.3498
17	0.7935	0.7209	0.8618	-0.0065	1	0.3353
18	0.8116	0.7319	0.8909	0.0116	1	0.3445
19	0.8463	0.7639	0.9243	0.0463	1	0.3343

rep	mean	q025	q975	bias	cover	accept
20	0.8321	0.7519	0.9032	0.0321	1	0.3259

The two non-covering replicates are useful diagnostics rather than failures. Replicate 6 estimates γ below the truth, whereas replicate 12 estimates it above the truth. This mixed direction supports the interpretation that the residual variation is replicate-specific rather than a systematic upward or downward bias.

Table 8: Replicates where the 95% posterior interval for gamma does not cover the truth.

rep	mean	q025	q975	bias	cover	accept
6	0.7089	0.6285	0.7852	-0.0911	0	0.3598
12	0.9001	0.8050	0.9792	0.1001	0	0.3436



Dashed line is the true common gamma.

Figure 3: Posterior intervals for learned common gamma across replicates.

The gamma plot shows moderate replicate-to-replicate variation. Some replicates estimate γ above 0.8, and some estimate it below 0.8. This is exactly the pattern expected under weak identification: the posterior is informative, but not as tightly anchored as the regression posterior. Importantly, there is no systematic collapse toward the upper boundary.

8 Dynamic residual-risk path recovery

The dynamic residual-risk path is a primary latent target. Since λ_{tj} enters the mean multiplicatively, path recovery is evaluated primarily on the log scale.

For each replicate, we summarize

$$\text{RMSE}_{\log \lambda} = \left[\frac{1}{Tn_1} \sum_{t=1}^T \sum_{j=1}^{n_1} \{\log \hat{\lambda}_{tj} - \log \lambda_{tj}^{\text{truth}}\}^2 \right]^{1/2}.$$

We also report 95% interval coverage for $\log \lambda_{tj}$ and the correlation between posterior mean and truth. Temporal increments,

$$\Delta \log \lambda_{tj} = \log \lambda_{tj} - \log \lambda_{t-1,j},$$

are included as a harder diagnostic because they ask whether the model captures short-term movement rather than broad latent-risk level.

Table 9: Dynamic residual-risk path recovery under learned gamma.

metric	value
Mean log lambda RMSE	0.0477
Mean log lambda MAE	0.0372
Mean log lambda 95% coverage	0.9301
Mean cor(log lambda)	0.7855
Mean RMSE of delta log lambda	0.0201
Mean cor(delta log lambda)	0.1847

The broad latent path recovery is good. The mean log-lambda RMSE is 0.0477, and the mean correlation between posterior mean and true $\log \lambda$ is 0.786. Coverage is 0.930, slightly below the nominal 0.95 target but still acceptable in this validation context.

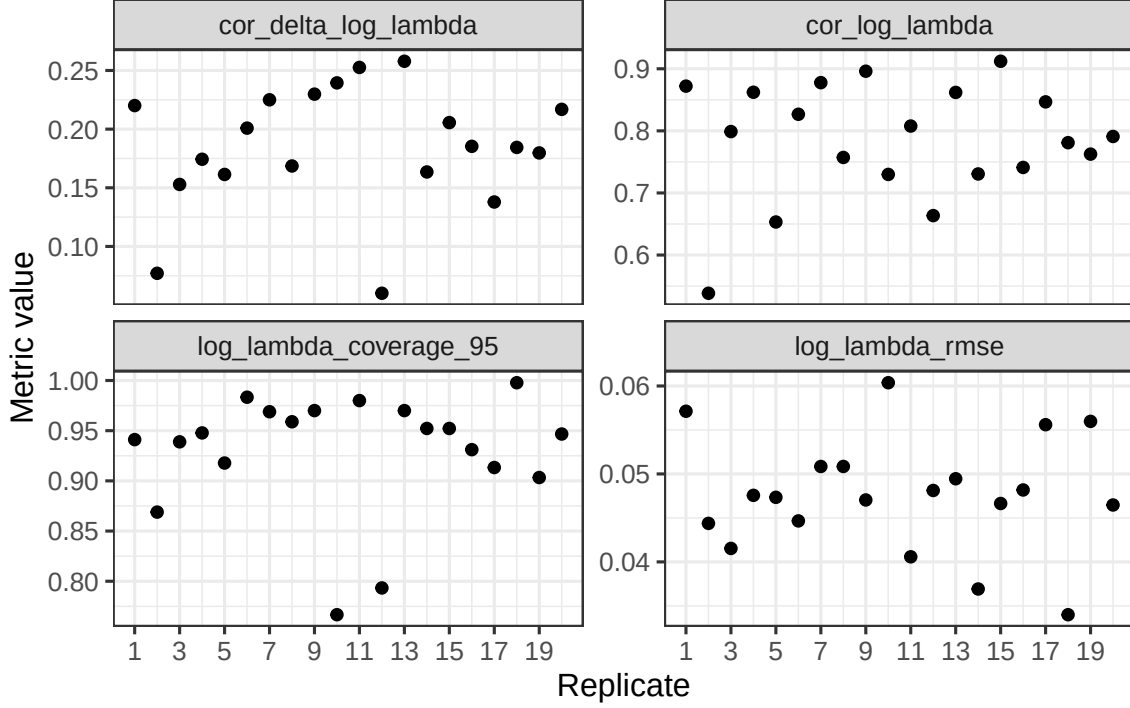


Figure 4: Latent dynamic risk recovery metrics by replicate.

The figure shows that log-lambda RMSE is consistently small across replicates. The level correlation is generally strong, although some replicates are harder than others. The increment correlation is much lower, which is expected. Recovering exact one-step changes is a much harder target than recovering the smooth level of a latent risk path.

9 Spatial and dispersion recovery

The spatial and dispersion components should remain stable when γ is learned. If learning γ created serious confounding with the rest of the model, we would expect degraded recovery of ϕ_j or r_j .

Table 10: Spatial and dispersion recovery summaries.

metric	value
Mean phi RMSE	0.0251
Mean phi correlation	0.9968
Mean r bias	0.4452
Median r bias	0.3555

metric	value
Mean r acceptance rate	0.3053

Spatial recovery is very stable. The mean spatial RMSE is 0.0251, and the mean correlation between posterior mean and true ϕ is 0.9968. These values indicate that the spatial effect is not being distorted by learning γ .

The dispersion estimates show mild positive bias on average, with mean bias 0.445. Relative to the true mean dispersion level of about 15, this bias is small and does not suggest a major dispersion failure.

10 MCMC diagnostics

The main MCMC diagnostics summarized in this report are acceptance rates for the spatial, dispersion, and gamma updates, rejection summaries for the beta elliptical slice step, and replicate-level summaries of gamma posterior behavior. External trace plots can be inspected separately when available, but the figures below are generated directly from the formal replicate summaries so that this report remains self-contained.

Table 11: MCMC diagnostic summaries across Scenario 3 replicates.

metric	value
Mean phi acceptance rate	0.2536
Mean r acceptance rate	0.3053
Mean gamma acceptance rate	0.3450
Mean beta ESS rejection count	14.4897
Mean elapsed seconds per replicate	299.4873

The MCMC diagnostics are stable. The mean spatial acceptance rate is 0.254, the mean dispersion acceptance rate is 0.305, and the mean gamma acceptance rate is 0.345. These values are in a reasonable range for random-walk Metropolis updates in this model.

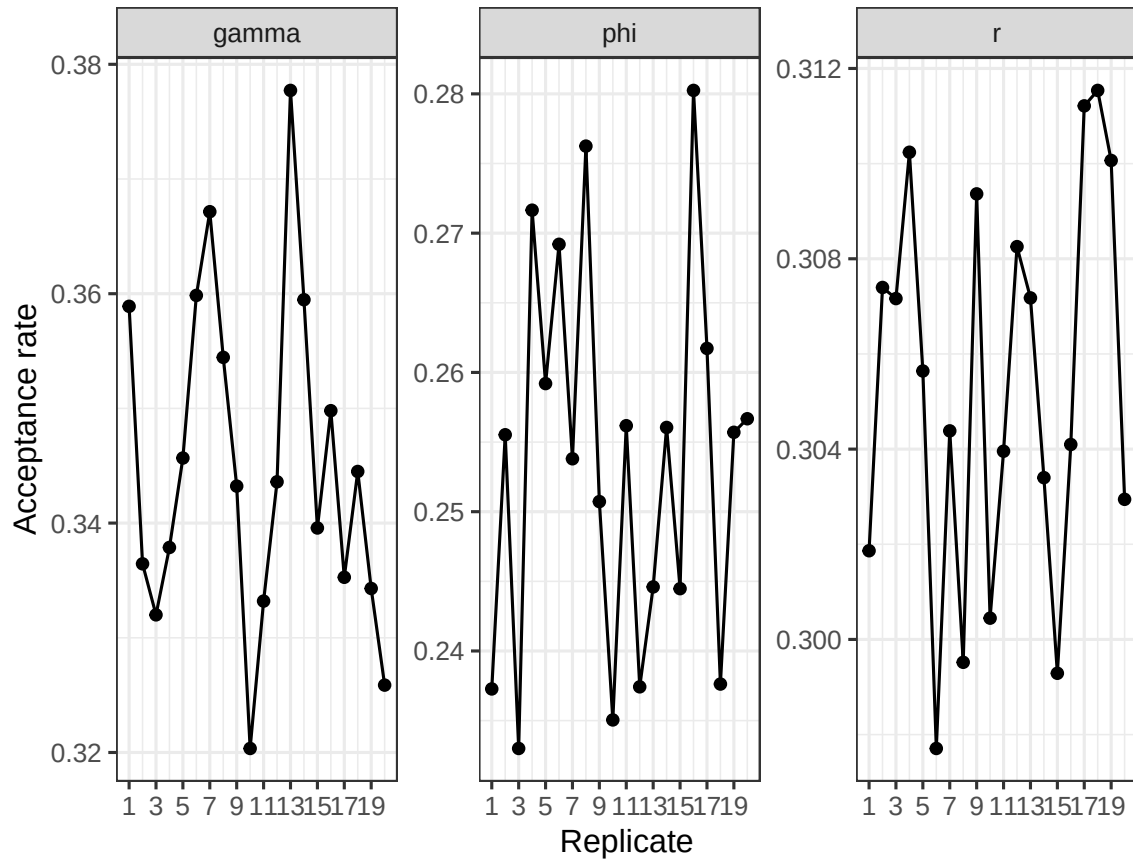


Figure 5: Scenario 3 acceptance rates by replicate.

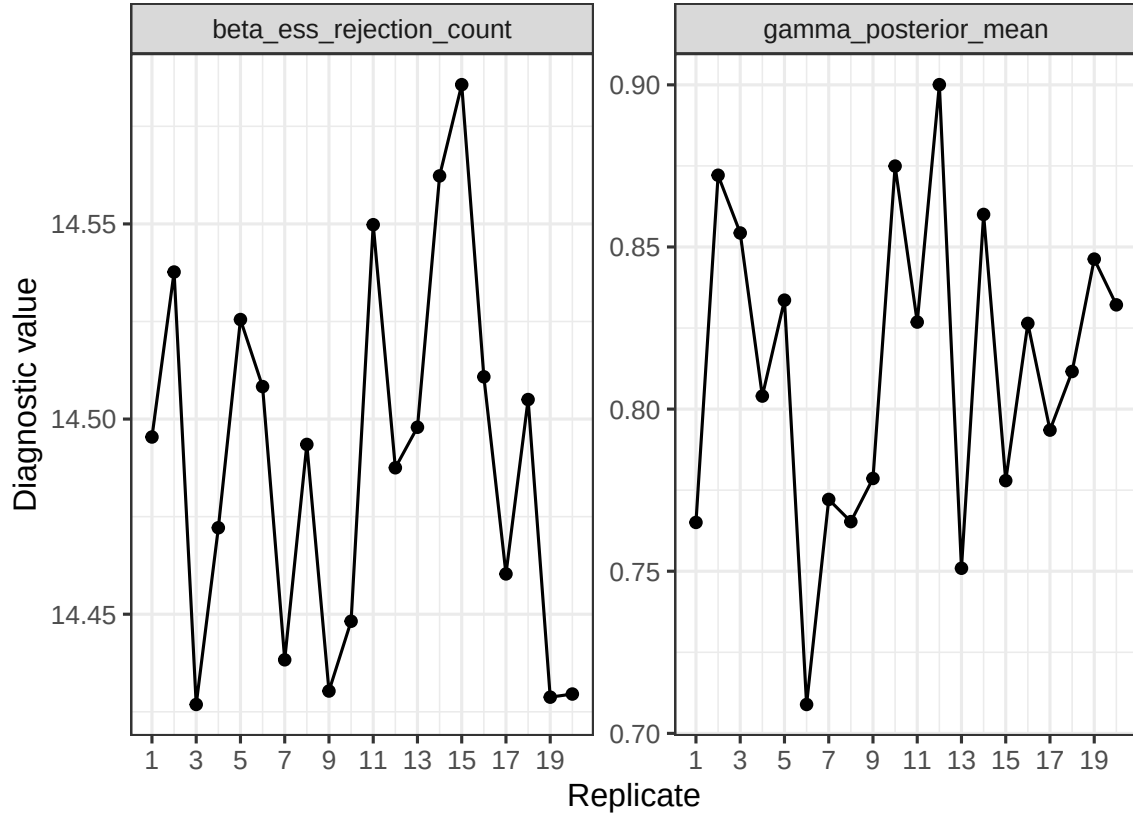


Figure 6: Scenario 3 gamma posterior mean and beta ESS rejection count by replicate.

The replicate-level diagnostics are intended to catch practical MCMC problems. The gamma update is especially important in Scenario 3 because γ is learned rather than fixed. The acceptance-rate figure shows that the spatial, dispersion, and gamma updates remain stable across replicates. The beta rejection-count diagnostic is also stable, suggesting that the regression block is not being disrupted by the additional gamma update.

Detailed trace and autocorrelation plots saved in `figures_mcmc/` can be inspected as an additional visual check. The summary figures included here focus on the replicate-level stability of the formal run and avoid relying on external image files.

11 Summary and interpretation

Scenario 3 validates the full dynamic model with learned common gamma. The results support five conclusions.

First, Scenario 2 and Scenario 3 are a clean comparison because both use the same DGP_DYNAMIC_T100 source data. This means the formal comparison isolates the fitting difference between fixed γ and learned γ .

Second, learning γ does not materially degrade regression recovery. The RMSE biases for β_1 and β_2 are 0.0121 and 0.0084, with coverage rates 0.95 and 1.00.

Third, using a common learned γ is appropriate for this scenario because the true data-generating mechanism uses a common γ . Estimating separate region-specific γ_j values would be a different and harder sensitivity analysis, not the main Scenario 3 validation.

Fourth, learned-gamma inference is stable but should be interpreted cautiously. The average posterior mean of γ is 0.813, close to the truth 0.80. The average bias is 0.013, the RMSE bias is 0.049, and coverage is 0.90. The slightly subnominal coverage is consistent with the fact that gamma is more weakly identified than the regression parameters.

Fifth, the dynamic risk path remains well recovered. The mean log-lambda RMSE is 0.0477, and the mean correlation between posterior mean and true $\log \lambda$ is 0.786. These results are nearly identical to Scenario 2, supporting the conclusion that learning the common temporal discount parameter does not compromise the main inferential targets.

Overall, Scenario 3 supports the staged validation strategy. Scenario 1 validates the baseline regression, spatial, and dispersion components. Scenario 2 validates dynamic residual-risk recovery when temporal smoothing is fixed. Scenario 3 shows that allowing the common temporal smoothing parameter to be learned still preserves the main model recovery properties. The correct interpretation is not that γ is always estimated perfectly, but that uncertainty in γ does not materially harm the scientific quantities of primary interest.